

Question Paper

Exam Date & Time: 21-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SUBJECT: PHY 1003/PHY_1003 APPLIED PHYSICS FOR ENGINEERS

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Illustrate the different types of optical fibres based on their refractive index profiles. Discuss the communication application of optical fiber with block diagram. (5)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 1B) An atom has two energy levels with a transition wavelength of 582 nm. At 300 K, 4.0×10^{20} atoms are in the lower state. (a) How many occupy the upper state under conditions of thermal equilibrium? (b) Suppose, instead, that 7.0×10^{20} atoms are pumped into upper state, with 4.0×10^{20} atoms in the lower state. How much energy could be released in a single laser pulse? (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) If a laser medium has a very short excited-state lifetime, how does it affect laser action? Justify your answer. (2)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2A) Analyze the relationship between the scattering angle and the wavelength shift when the target metal is exposed to extremely intense and monochromatic electromagnetic radiation and discuss the physical significance of the Compton effect. (5)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2B) In an experiment, the tungsten cathode which has a threshold of 435 nm is irradiated by ultraviolet light of wavelength 215 nm. Calculate (i) maximum energy of emitted photoelectrons, and (ii) work function for tungsten (in eV). (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2C) If the frequency of incident radiation is increased beyond the threshold while intensity is kept constant, what happens to:
(i) The number of emitted electrons, and
(ii) Their kinetic energy? (2)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3A) Solve the Schrodinger equation for a particle of mass m in a one-dimensional infinite potential well (box) of length L and obtain an expression for wavefunction of the particle. (5)

Course Outcome: "CO3" / Cognitive Level: "Understand"

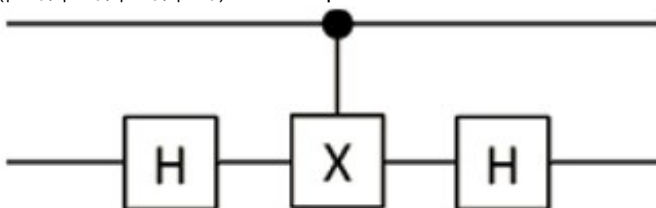
- 3B) A student conducted an experiment on the photoelectric effect by shining light with a wavelength of 500 nm onto a metal surface. To stop the flow of photocurrent, the student applied a negative potential until the current dropped to zero. He repeated the experiment by varying the distance of the light source (with no other changes made to the experimental setup). How do these variations in distance affect the stopping potential and photocurrent? Justify your answer. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- 3C) Well-1 and Well-2 are two square wells having same length. Well-1 has walls of finite height, and well-2 has walls of infinite height. Both the wells contain identical quantum particles, one in each well. Compare the magnitude of the ground-state momentum. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4A) For the quantum circuit shown below, identify the output for each of the computational basis states ($|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$) as the input. (5)



Based on the obtained output, draw the equivalent two-qubit gate for the above quantum circuit.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4B) Explain Bloch sphere representation of a qubit. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4C) **Both the set of all complex column matrices of order $(n \times 1)$ and the set of all complex row matrices of order $(1 \times n)$ form linear vector spaces (LVS) over the field of complex numbers.** State whether the given statement is True or False. Justify your answer. (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5A) Explain the postulates of Quantum Mechanics. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5B) Show that the SWAP gate/circuit interchanges the state of the two qubits. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5C) **Given:** (2)

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \quad |\psi'\rangle = \frac{i}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle$$

Are $|\psi\rangle$ and $|\psi'\rangle$ globally phase equivalent? Explain.

Course Outcome: "CO4" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 PHY_1003:APPLIED PHYSICS FOR ENGINEERS

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	7
4	4	CO 4	20
5	5	CO 5	3
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 PHY_1003:APPLIED PHYSICS FOR ENGINEERS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	10(20%)
3	Application	10	33(66%)
4	Analysis	3	7(14%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)